# Morphology of droplets in turbulent flows

New Jersey Institute of Technology, 2023 November 27

**Ianto Cannon**, Giovanni Soligo, and Marco E. Rosti
Okinawa Institute of Science and Technology



# We care about droplet morphology

$CO_2$ absorption[a]

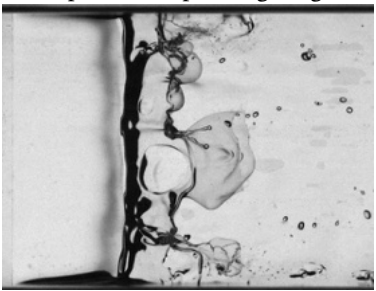


Wave breaking on the coast of Okinawa[b].

[a] Merlivat, L., & Memery, L. (1983). *J. Geophys. Res.-Oceans*, *88*(C1), 707–724.

[b] Photo credit: Linda Gietka

Droplet breakup during cough[a]



1mm water & glycerol film with 15m/s airflow.

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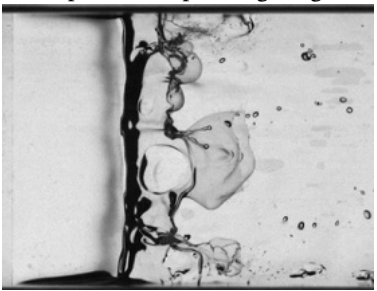


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What is the morphology of droplets in turbulence?

How do surfactants affect morphology?

# Numerical method

We use the in-house solver *Fujin*.



Fluid motion,

$$\rho\frac{\partial \mathbf{u}}{\partial t} + \rho\mathbf{u}\cdot\nabla\mathbf{u} = -\nabla p + \nabla\cdot\left[\mu\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right)\right] + \nabla\cdot(\boldsymbol{\tau}_c f_\sigma) + \mathbf{f}_s,$$

$$\nabla\cdot\mathbf{u} = 0, \tag{1}$$

Droplet motion[a],

$$\frac{\partial \phi}{\partial t} + \nabla\cdot(\mathbf{u}\phi) = 0, \tag{2}$$

Surfactant motion[b],

$$\frac{\partial \psi}{\partial t} + \mathbf{u}\cdot\nabla\psi = \nabla\cdot(M_\psi\nabla\mu_\psi), \tag{3}$$

Surface tension[c],

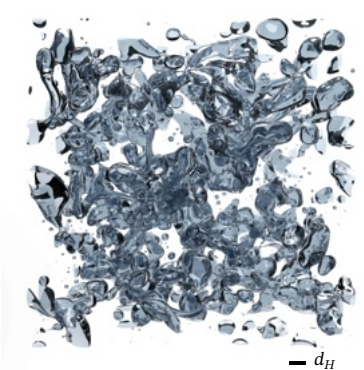
$$f_\sigma = \max[\sigma_{min}, \sigma_0(1+\beta_s\log(1-\psi))]. \tag{4}$$

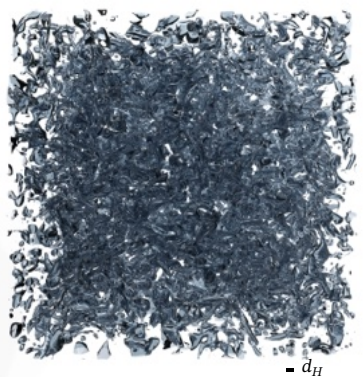
[a] Ii, S., Sugiyama, K., Takeuchi, S., Takagi, S., Matsumoto, Y., & Xiao, F. (2012). *Journal of Computational Physics*, *231*(5), 2328–2358.
[b] Engblom, S., Do-Quang, M., Amberg, G., & Tornberg, A. (2013). *Commun. Comput. Phys.*, *14*(4), 879–915.
[c] Muradoglu, M., & Tryggvason, G. (2008). *Journal of Computational Physics*, *227*(4), 2238–2262.

# Simulation setup

- Tri-periodic domain with $500 \times 500 \times 500$ grid points.
- Six simulations with three values of surface tension, and two surfactant concentrations.
- Hinze-scale[d] $d_H \equiv 0.725\sigma^{3/5}\rho^{-3/5}\epsilon^{-2/5}$.

| case | | $\langle\phi\rangle$ | $\langle\psi\rangle$ | $We_e$ | $d_H/L$ |
|---|---|---|---|---|---|
| C10 | ● | 0.1 | 0 | **9.91** | **0.0560** |
| C20 | ● | 0.1 | 0 | 20.0 | 0.0373 |
| C40 | ● | 0.1 | 0 | 39.3 | 0.0252 |
| S05 | ◆ | 0.1 | 0.1 | 12.7 | 0.0489 |
| S10 | ◆ | 0.1 | 0.1 | 25.8 | 0.0338 |
| S20 | ◆ | 0.1 | 0.1 | **49.5** | **0.0232** |



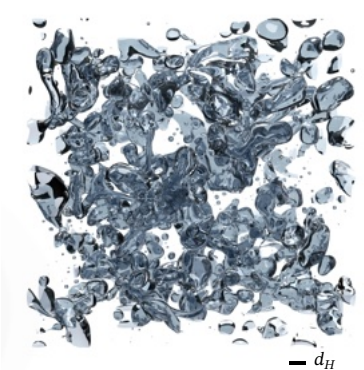



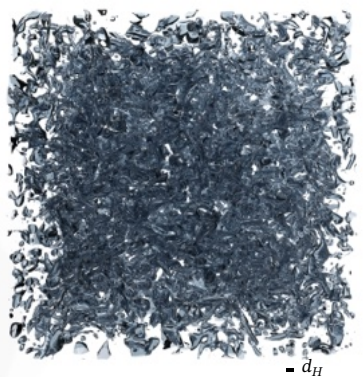

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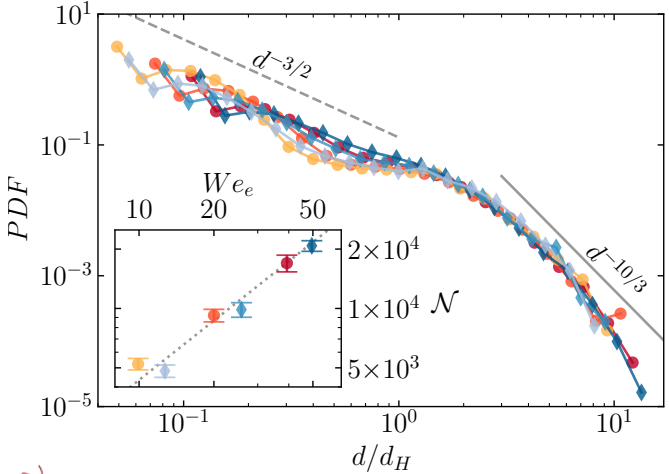
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# Counting droplets.



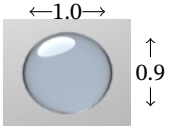

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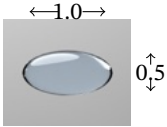
- We use a flood-fill algorithm to count the droplets[a].
- Surface tension $\sigma \downarrow$, number of droplets $\mathcal{N} \uparrow$.
- Define a volume-based diameter for each droplet, $d \equiv (6V/\pi)^{1/3}$.
- $d < d_H$: coalescence.
- $d > d_H$: breakup[b].

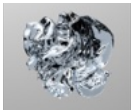
[a] Newman, W. M., & Sproull, R. F. (Eds.). (1979). McGraw-Hill, Inc.

[b] Deane, G. B., & Stokes, M. D. (2002). *Nature*, *418*(6900), 839–844.

# What shape are the droplets?



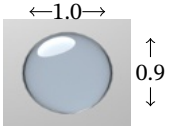



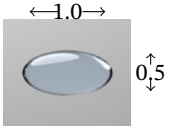



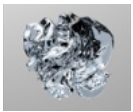
| | | | |
|---|---|---|---|
| Aspect ratio $\sqrt{I_1/I_3}$ | 0.946 | 0.633 | 0.905 |
| Sphericity $d/d_A$ | 0.999 | 0.965 | 0.701 |

- $I_1$ and $I_3$ are the smallest and largest moments of inertia.
- Area-based diameter for each droplet, $d \equiv \sqrt{A/\pi}$.

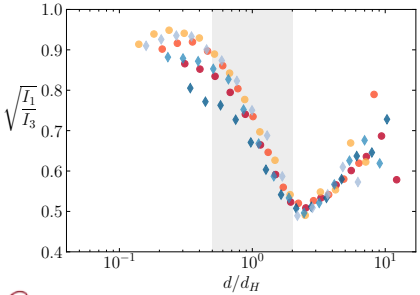
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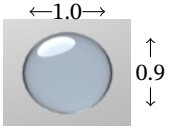



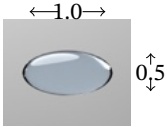
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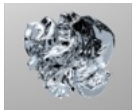



- $I_1$ and $I_3$ are the smallest and largest moments of inertia.
- Area-based diameter for each droplet, $d \equiv \sqrt{A/\pi}$.
- $d << d_H$ aspect ratio $\to 1$.
- $d \approx d_H$ droplets are extended.
- $d >> d_H$ aspect ratio $\to 1$.

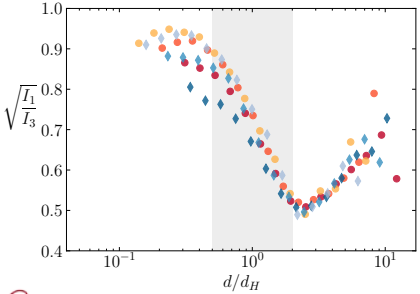
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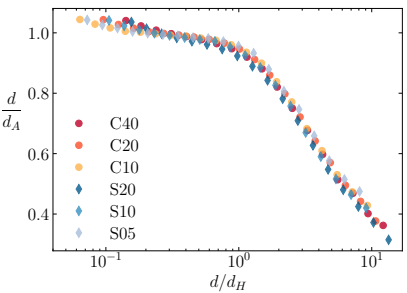


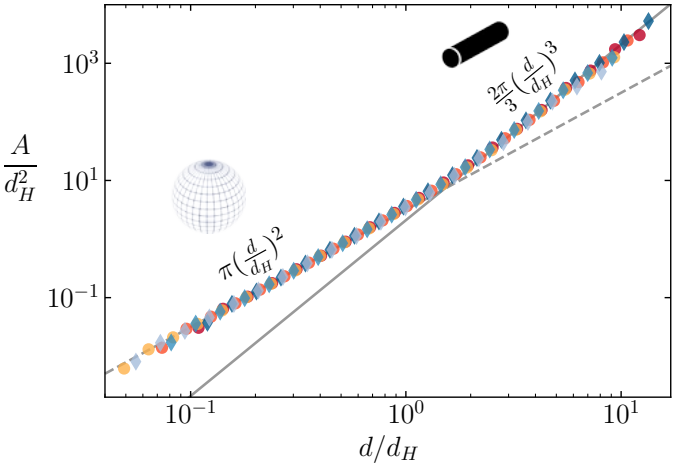
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- $d > d_H$ area $\uparrow$.

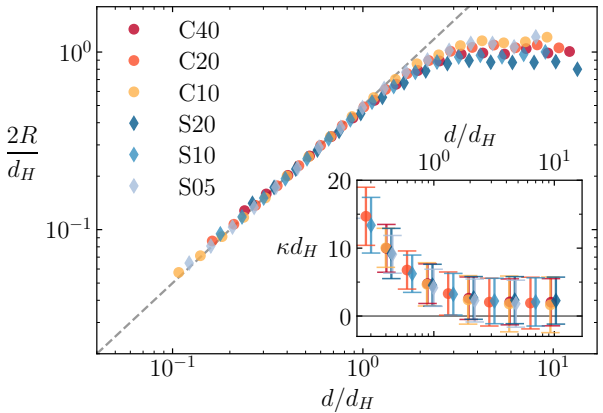
# What is the area of the droplets?

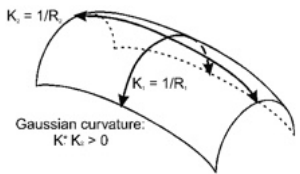



- $d < d_H$: spheroids.
- $d > d_H$: filaments with diameter $d_H$.

# Interface curvature

Curvature $\kappa = \kappa_1 + \kappa_2 = 1/R$.

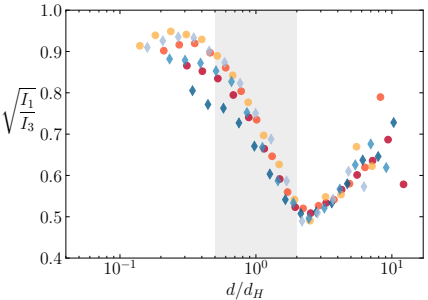



- $d < d_H$: $R = d/4$.
- $d > d_H$: $R = d_H/2$, with regions of negative curvature.

# Recap

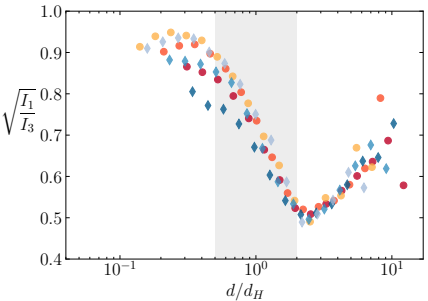
- The large droplets are filamentous.



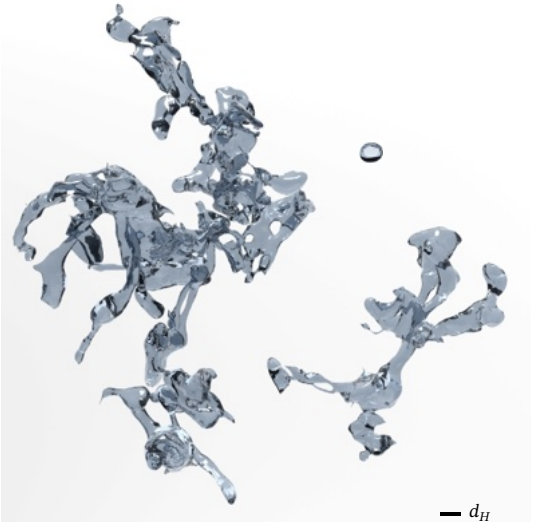

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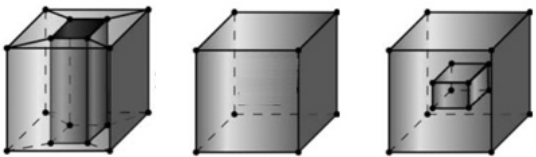

- Aspect ratio → 1 implies the filaments must be coiled.



## How to count the number of handles?

We compute the Euler characteristic[e] of the droplet interfaces.

$$\chi = n - e + f \tag{5}$$



$\chi = 0$ $\chi = 2$ $\chi = 4$

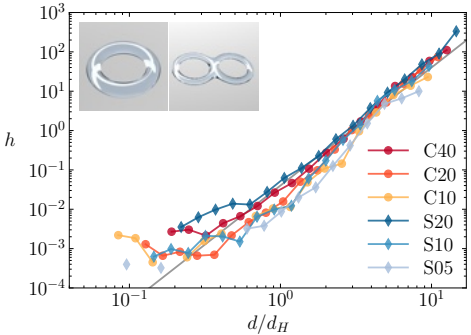
This gives the number of handles[f] $h$ and voids $v$.

$$\chi = 2 + 2v - 2h. \tag{6}$$

[e] Massey, W. S. (1997). Springer.

[f] Mendoza, R., Thornton, K., Savin, I., & Voorhees, P. (2006). *Acta Mater.*, *54*(3), 743–750.

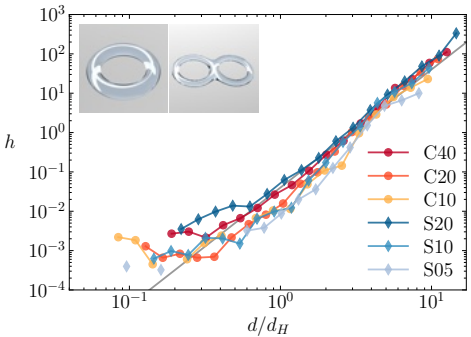
# How many handles and voids are there?




$$h = 0.04(d/d_H)^3$$

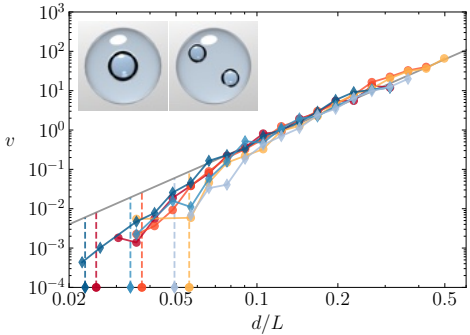
- Constant number of handles per unit length of filament.

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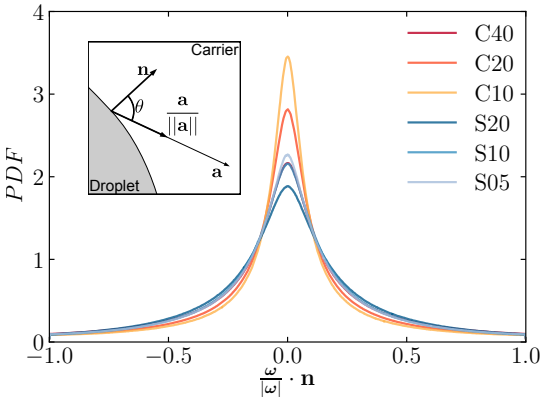
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$v = 500(d/L)^3$

- Constant number of voids per unit volume of droplet.

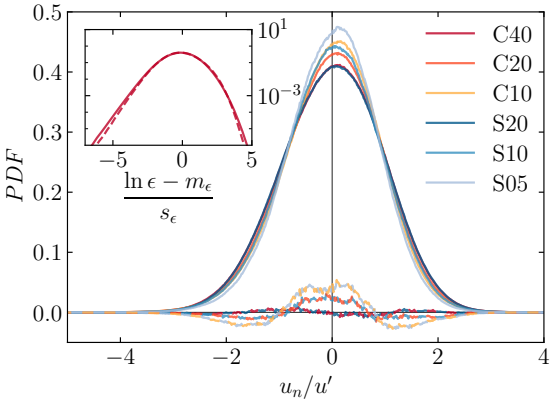
# Vorticity at the interface




- Vorticity $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$.
- Normal to the interface $\boldsymbol{n}$.
- Vorticity is mostly tangential to the interface[a].
- Surface tension ↓, effect ↓.
- Surfactants increase velocity gradients tangential direction.

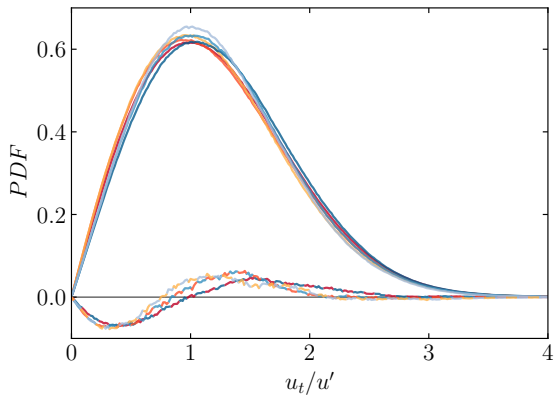
[a] Shao, C., Luo, K., Yang, Y., & Fan, J. (2018). *International Journal of Multiphase Flow, 107*, 263–274.

# Flow normal to interface




- Extreme inward velocities are more likely than outward.
- $m_\epsilon$ and $s_\epsilon$ are the mean and standard deviation of the dissipation $\epsilon$.
- Surfactant suppresses large normal velocities.

# Flow tangential to interface




- Surfactant induces large tangential velocities.

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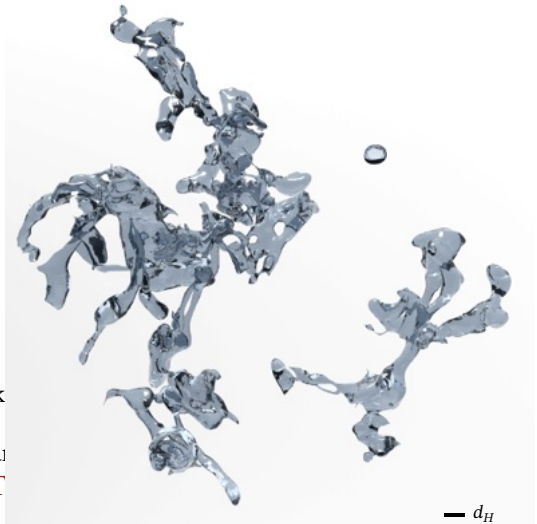
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[a] Cannon, I., Soligo, G., & Rosti, M. E. (2023). *arXiv:2307.15448*.

# Bibliography I


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